

The empirical recurrence for OEIS A223998, recovered from its tail

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Abstract

OEIS A223998 records “Empirical recurrence of order 54” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 239$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 62$. Its last coefficient is $c_{54} = 16782458880 \neq 0$, so no further tail satisfies anything shorter; any order-54 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A223998 is “Number of $n \times 7$ 0..2 arrays with antidiagonals unimodal and rows and diagonals nondecreasing.”. It has offset 1 and begins

36, 736, 8291, 71061, 559188, 4368458, 34534687, 276286000, ...

The entry states:

Empirical recurrence of order 54 (see link above)

As of the “Last modified” line on the live entry (Oct 03 2025 23:10:02, revision 10) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 239$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 239$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 54: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 54.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 478$ exact terms from $n = 8$ on, it returns order 54 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{54} c_i a(n-i),$$

c_1	36	c_{15}	−172309723	c_{29}	5933241933746	c_{43}	120540644103512
c_2	−525	c_{16}	302595704	c_{30}	−8716725460315	c_{44}	−23006882415792
c_3	3796	c_{17}	−181536213	c_{31}	1054557878608	c_{45}	73738982853248
c_4	−11714	c_{18}	−1762464511	c_{32}	10852477794434	c_{46}	−11657384182208
c_5	−13477	c_{19}	9366044740	c_{33}	11443576645052	c_{47}	47460137557888
c_6	194361	c_{20}	−10257303825	c_{34}	−4434875780290	c_{48}	−7885572753920
c_7	−274084	c_{21}	−16375743838	c_{35}	26223899166584	c_{49}	12492028701184
c_8	−1414596	c_{22}	2498483684	c_{36}	4854304614752	c_{50}	−506542140416
c_9	4650220	c_{23}	137064498056	c_{37}	59351448765460	c_{51}	2423206543360
c_{10}	1644589	c_{24}	−245678493438	c_{38}	90065735990524	c_{52}	−89554944000
c_{11}	−21283228	c_{25}	187935480986	c_{39}	183169819971716	c_{53}	733870817280
c_{12}	3237545	c_{26}	−286682456811	c_{40}	71694284624536	c_{54}	16782458880
c_{13}	77424978	c_{27}	594175351787	c_{41}	117207837077480		
c_{14}	−52139474	c_{28}	−1077419487703	c_{42}	−6361196609640		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 62$, and it is the recurrence OEIS A223998 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{54} - c_1x^{54-1} - \dots - c_{54}$. Its constant term is $-c_{54} = -16782458880$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 54 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 54, so $\deg q = 54$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 54, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 54, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 16782458880.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-54 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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